

PROJECT DOCUMENTATION:

HERITAGE GUARD

Securing the Past for the Future: A Digital Platform for Artifact Documentation and Chain of Custody

Program: FIRST LEGO League (FLL) Challenge

Season Theme: UNEARTHED

Project Domain: Cultural Heritage Preservation & Digital Asset Management

1. EXECUTIVE SUMMARY

Heritage Guard is a secure, role-based web platform engineered to modernize how archaeological teams and museums document, track, and protect historical artifacts. Aligned with the FIRST LEGO League *UNEARTHED* challenge, this project bridges the gap between historical preservation and modern digital security. By replacing fragmented paper trails with a centralized digital system, Heritage Guard ensures a verifiable chain of custody from the moment an artifact is unearthed to its final placement in a museum gallery, effectively combating theft, transport damage, and administrative mismanagement.

2. AREA OF FOCUS

This project operates at the intersection of Cultural Heritage Preservation, Archaeological Data Management, and Asset Security. While traditional archaeological processes focus heavily on the physical excavation of items, Heritage Guard targets the highly vulnerable post-discovery phase. Our focus is to secure the data and physical location of artifacts, ensuring that once an item is unearthed, its historical integrity and physical safety are permanently protected through digital oversight.

3. PROBLEM STATEMENT

Archaeological teams and museums currently struggle with fragmented, outdated, and inefficient methods for managing artifact inventories. The core problems facing these institutions include:

- **Theft and Missing Items:** High-value artifacts are frequently stolen from active dig sites or museum storage facilities. Without real-time tracking, these thefts often go unnoticed until periodic, delayed inventory audits are conducted.
- **Transport Damage:** Artifacts suffer irreversible damage—such as cracks, surface erosion, or structural breaks—when moved between dig sites, conservation labs, and display galleries, largely due to improper handling records and miscommunication between teams.
- **Lost Items in Storage:** Unclear location tracking causes items to be misplaced in vast storage facilities, effectively losing them to researchers and the public for decades.
- **Fragmented Record-Keeping:** When multiple individuals (archaeologists, conservators, security staff) handle the same object, disconnected spreadsheets and paper logs lead to a broken chain of custody, making it impossible to hold individuals accountable for an artifact's safety.

4. PROBLEM BEING SOLVED (THE SOLUTION)

Heritage Guard solves these challenges by transforming everyday documentation into a clear, shared, and secure digital system. The platform addresses the lack of accountability and real-time visibility by digitizing the entire lifecycle of an artifact. It replaces slow, error-prone manual logs with an automated, trackable, and auditable digital footprint, ensuring that every stakeholder knows exactly where an artifact is, what condition it is in, and who last handled it.

5. KEY FEATURES AND THEIR USES

Heritage Guard is equipped with specialized tools designed to address specific operational needs within archaeological and museum environments:

- Unique Artifact Codes:
 - *Feature:* Auto-generation of standardized alphanumeric codes (e.g., HG-1001) for every new entry.
 - *Use:* Allows field workers to instantly tag and identify items without relying on confusing local naming conventions.
- Real-Time Status Tracking:
 - *Feature:* Artifacts are categorized dynamically as *OK*, *Broken*, or *Stolen*, triggering visual alerts on the dashboard.
 - *Use:* Enables security staff and managers to instantly see which items require urgent attention, replacing the need for manual check-ins.
- Digital Chain of Custody (Movement Log):
 - *Feature:* A timestamped, uneditable history of every location change and status update, recording the user who made the change.
 - *Use:* Provides legal and administrative accountability. If an item goes missing, administrators know exactly who handled it last and where it was supposed to be.
- Structured Condition Checklists:
 - *Feature:* Interactive digital checklists (e.g., "Crack on surface," "Missing fragment") that appear when an item is marked as damaged.
 - *Use:* Ensures damage is reported consistently and professionally, giving conservators exact details on what needs to be repaired.
- High-Value Watchlist:
 - *Feature:* A dedicated tracking system where fragile or highly valuable items can be flagged with a single click.
 - *Use:* Surfaces critical items to the top of the dashboard, ensuring they receive priority monitoring and handling.
- Automated Reporting & Activity Logs:

- *Feature:* Printable summaries of problem items, location distributions, and a complete feed of user activity.
- *Use:* Allows museum directors and site managers to generate audit reports for stakeholders or law enforcement instantly.

6. UI & UX DESIGN

The User Interface (UI) and User Experience (UX) of Heritage Guard are built around the philosophy of "Professional Heritage meets Modern Dashboarding."

- **Visual Identity:** The platform utilizes a "Heritage Theme" palette consisting of Terracotta, Gold, and Parchment tones, evoking a sense of history and antiquity. This is paired with modern, clean typography (using the *Cinzel* serif font for headers and *Inter* for body text) to maintain a highly professional and legible appearance.
- **Light/Dark Mode:** A seamlessly integrated theme toggle allows users to switch between light and dark modes. Dark mode is heavily utilized during late-night lab work or field work in low-light environments to reduce eye strain.
- **Responsive Grid Layout:** Built with modern CSS Grid, the dashboard and artifact pages automatically scale to fit any screen size—from a desktop monitor in a museum office to a tablet used in an excavation trench.
- **Visual Cues & Micro-interactions:** Status badges are color-coded (Green for OK, Amber for Broken, Red for Stolen) to allow for instant visual scanning. Subtle animations, such as fade-ins and card hover lifts, provide a fluid user experience without sacrificing load speed.
- **Role-Based Dashboards:** The UX adapts to the user. A *Trial User* sees a clean, read-only interface, while an *Archaeologist* sees prominent "Add New Artifact" buttons and editing tools.

7. HOW IT WORKS (SYSTEM WORKFLOW)

Heritage Guard operates through a simple but highly secure, role-based workflow:

1. **Authentication & Role Assignment:** Users must register or log in. Upon registration, they select their specific role (Archaeologist, Museum Worker, Security, or Trial User). The system dynamically grants permissions based on this role.
2. **Artifact Registration:** An archaeologist unearths an item, logs into the system, and creates a new entry. They input the item's name, era, initial location, and notes. The system generates the unique tracking code (e.g., HG-1006).
3. **Tracking & Movement:** As the artifact moves from the dig site to a museum lab, the Museum Worker updates the location field. The system automatically logs this movement in the artifact's history file with a timestamp and the user's identity.
4. **Condition Updates:** If the item is damaged during transport, the user changes the status to "Broken." The UI instantly presents a condition checklist. The user selects the specific damages, which are then saved to the artifact's profile and flagged on the main dashboard.
5. **Monitoring & Reporting:** Security staff monitor the main dashboard for red/stolen alerts. Site managers can navigate to the "Reports" page to print a physical PDF of all broken or stolen items for administrative filing.

8. MAIN PEOPLE IT IS SERVING (BENEFICIARIES)

While the platform is operated by museum and field staff, the ultimate beneficiaries of Heritage Guard are future generations, academic researchers, and the global public. By preventing the loss, theft, or destruction of historical items, we ensure that cultural heritage remains intact for study, education, and exhibition. We are serving the artifacts themselves—giving them a digital shield that guarantees their preservation from the moment they are uncovered.

9. USER TARGET (PLATFORM USERS)

Heritage Guard utilizes a Role-Based Access Control (RBAC) system, ensuring that different team members have the specific tools they need without compromising system security. The primary users of the platform are:

1. Archaeologists: Utilize the platform in the field to register new finds, document initial observations, and assign unique tracking codes before items leave the excavation site.
2. Museum Workers & Conservators: Update artifact locations as items move into labs or galleries, log transportation damage, and fill out condition checklists during the restoration process.
3. Security Staff: Monitor the dashboard for active alerts, quickly flag items as "Stolen" during security breaches, and track the movement history of high-value items on the watchlist.
4. Trial Users (Guests): Granted read-only access to the artifact registry, allowing students, auditors, or stakeholders to view the system's capabilities without altering sensitive data.

10. MARKET TARGET

Heritage Guard is designed for scalability and targets specific institutional markets within the cultural and historical sectors:

- Academic Institutions & Universities: Departments conducting seasonal field excavations that require organized, transferable digital records to bring back to their respective labs.
- Local and Regional Museums: Institutions that may lack the budget for enterprise-level collection management software but require robust tracking for smaller inventories and rotating exhibits.
- Government Cultural Ministries: State bodies overseeing historical sites and national storage facilities that require strict chain-of-custody documentation to prevent the illicit trafficking of antiquities.
- Private Archaeological Contractors: Independent firms hired for infrastructure-adjacent excavations needing to prove compliance and safe handling of recovered artifacts to government regulators.

11. CONNECTION TO THE UNEARTHED SEASON

The FIRST LEGO League *UNEARTHED* season challenges teams to explore archaeology and help uncover the past. Heritage Guard deeply connects to this mission by recognizing that unearthing an artifact is only the first step; protecting it is the true challenge. By providing a modern, digital infrastructure, Heritage Guard ensures that what is found can be studied, shared, and preserved for the future, rather than being lost, damaged, or stolen in the chaotic aftermath of discovery.

12. CONCLUSION

Heritage Guard bridges the gap between historical preservation and modern digital security. By addressing the specific pain points of theft, transport damage, and fragmented record-keeping, the platform empowers archaeologists, museum workers, and security teams to work collaboratively. Through unique artifact codes, real-time status tracking, a thoughtful user interface, and an unbreakable digital chain of custody, Heritage Guard turns everyday documentation into a clear, shared system—ensuring that the legacy of our past is safely secured for the future.